

# **RESEARCH PROTOCOL OUTLINE (Version 0.1)**

## **1. Project Title**

### **Proposed Title**

Digital Transformation and Its Impact on the Quality and Efficiency of Higher Education: A Case Study of the Universidad Nacional Mayor de San Marcos

### **Keywords**

- Digital Transformation
- Higher Education
- Educational Quality
- Institutional Efficiency
- Public Universities
- Organizational Innovation
- University Governance
- Peru

## **2. Problem Statement**

### **Background**

Higher education institutions worldwide are experiencing profound organizational changes driven by digital transformation. Universities are progressively integrating digital technologies into academic management, teaching, research, student services, and institutional governance. Digital transformation is no longer limited to technological modernization but has become a strategic process capable of reshaping institutional performance and competitiveness.

Although Peruvian public universities have implemented numerous digital initiatives, empirical evidence regarding their actual contribution to educational quality and institutional efficiency remains scarce. Most institutional efforts have focused on technological implementation rather than evaluating measurable organizational outcomes.

The Universidad Nacional Mayor de San Marcos (UNMSM), as Peru's oldest and most prestigious public university, represents an appropriate context for investigating how digital transformation influences institutional performance. Understanding this relationship is essential for supporting evidence-based decision-making and improving university governance.

## **Research Gap**

Existing literature primarily addresses digital transformation conceptually or through descriptive case studies. Few empirical investigations have simultaneously examine:

- Digital transformation maturity
- Educational quality
- Institutional efficiency
- Organizational mechanisms
- Public universities in emerging economies

This study seeks to address this gap.

## **3. Research Questions**

### **Main Research Question**

To what extent does digital transformation influence the quality and efficiency of higher education at the Universidad Nacional Mayor de San Marcos?

### **Supporting Research Questions**

**RQ1.** Which dimensions of digital transformation significantly influence educational quality?

**RQ2.** How does digital transformation affect institutional efficiency?

**RQ3.** Which organizational, technological, and human factors facilitate or hinder digital transformation?

**RQ4.** What strategic recommendations can strengthen digital transformation in order to improve educational quality and institutional efficiency?

## **4. Research Objectives**

### **General Objective**

To analyze the influence of digital transformation on the quality and efficiency of higher education at the Universidad Nacional Mayor de San Marcos.

### **Specific Objectives**

1. To evaluate the level of digital transformation within UNMSM.
3. To determine the relationship between digital transformation and educational quality.
4. To examine the influence of digital transformation on institutional efficiency.
5. To identify organizational and technological factors that facilitate digital transformation.

6. To propose strategic recommendations for strengthening institutional digital transformation.

## **5. Literature Review Summary**

The literature indicates that digital transformation extends beyond technological adoption and involves organizational change, innovation, leadership, governance, and digital capabilities.

Recent studies identify several critical dimensions:

- Digital Leadership
- Digital Governance
- Organizational Agility
- Digital Culture
- Technological Infrastructure
- Digital Competencies
- Student Experience
- Academic Innovation

Existing research demonstrates positive associations between digital transformation and:

- Educational quality
- Student satisfaction
- Institutional resilience
- Administrative efficiency

However, empirical studies focusing on public universities in Latin America remain limited.

Therefore, this research contributes by providing evidence from the UNMSM context.

## **6. Quasi-Experimental Design**

The study adopts a **Mixed-Methods Case Study** supported by a **quasi-experimental explanatory design**.

Two naturally occurring comparison groups will be considered.

Example

High Digital Transformation Units vs Low Digital Transformation Units

The study will compare indicators such as:

- Educational quality
- Student satisfaction

- Administrative efficiency
- Academic productivity

Quantitative analyses will be complemented by qualitative interviews and institutional document analysis.

The integration of both datasets will provide a comprehensive understanding of digital transformation outcomes.

## **7. Data Management Plan Outline**

The project will follow international best practices for research data management.

### **Data Sources**

#### Primary Data

- Surveys
- Interviews
- Focus groups

#### Secondary Data

- Institutional reports
- Strategic plans
- Digital transformation policies
- Academic indicators

### **Storage**

Encrypted institutional storage

Password-protected devices

Cloud backup

### **Data Quality**

Data cleaning

Coding procedures

Missing value analysis

Validation

### **Data Sharing**

Anonymized datasets

Research repository

Institutional archive

## **8. Ethical Considerations**

The study will comply with national and international ethical standards.

Key ethical principles include

- Voluntary participation
- Informed consent
- Confidentiality
- Data privacy
- Non-maleficence
- Research integrity

Participants' identities will remain anonymous.

Institutional authorization will be obtained before data collection.

Ethical approval will be requested from the corresponding Research Ethics Committee.

## 9. Project Schedule / Timeline

| Phase   | Activities               | Months |
|---------|--------------------------|--------|
| Phase 1 | Literature Review        | 1–2    |
| Phase 2 | Instrument Design        | 3      |
| Phase 3 | Pilot Study              | 4      |
| Phase 4 | Data Collection          | 5–6    |
| Phase 5 | Data Analysis            | 7–8    |
| Phase 6 | Integration of Findings  | 9      |
| Phase 7 | Thesis Writing           | 10–11  |
| Phase 8 | Final Review and Defense | 12     |

## 10. Expected Results and Impact

### Scientific Contributions

- Development of empirical evidence regarding digital transformation in higher education.
- Validation of a methodological framework integrating educational quality and institutional efficiency.
- Contribution to digital transformation literature in emerging economies.

### Practical Contributions

Evidence-based recommendations for university decision-makers.

Improved digital governance strategies.

Optimization of institutional processes.

Support for strategic planning.

### **Social Contributions**

- Improved student experience.
- Strengthened educational quality.
- More efficient public universities.
- Contribution to sustainable digital transformation in Peruvian higher education.

### **Expected Academic Contribution**

This research is expected to provide one of the first comprehensive empirical analyses of the relationship between digital transformation, educational quality, and institutional efficiency within a leading public university in Peru. By integrating quantitative and qualitative evidence under a pragmatic mixed-methods approach, the study will contribute to both the theoretical advancement of digital transformation research and the practical development of evidence-based strategies for higher education governance. The proposed methodological framework is intended to be transferable to other public universities in Latin America facing similar digital transformation challenges.